Started on	Monday, 28 September 2026, 8:20 PM
State	Finished
Completed on	Monday, 28 September 2026, 8:29 PM
Time taken	8 mins 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void merge(int a[], int left, int mid, int right) {
    int i = left;
    int j = mid + 1;
    int k = 0;

    int temp[right - left + 1];

    while (i <= mid && j <= right) {
        if (a[i] <= a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }

    while (i <= mid)
        temp[k++] = a[i++];

    while (j <= right)
        temp[k++] = a[j++];

    for (i = left, k = 0; i <= right; i++, k++)
        a[i] = temp[k];
}

void mergeSort(int a[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;

        mergeSort(a, left, mid);
        mergeSort(a, mid + 1, right);

        merge(a, left, mid, right);
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    mergeSort(arr, 0, n - 1);

    long long sum = 0;

    for (int i = 0; i < n; i++)
```

```
52 |         sum += (long long)arr[i] * i;
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.